

Sec: NAITS

Date: 15-02-2024

Time: 3 Hrs.

PTM – 02

Max. Marks: 300

PHYSICS

Section	Question Type	+Ve Marks	–Ve Marks	No. of Qs.	Total marks
Sec – I (Q.N : 1 – 20)	Questions with Single Correct Choice	4	–1	20	80
Sec–II (Q.N : 21 – 30)	Questions with Numerical Answer Type (+ / –Number)	4	–1	10	20
Total				30	100

CHEMISTRY

Section	Question Type	+Ve Marks	–Ve Marks	No. of Qs.	Total marks
Sec – I (Q.N : 31 – 50)	Questions with Single Correct Choice	4	–1	20	80
Sec–II (Q.N : 51 – 60)	Questions with Numerical Answer Type (+ / – Number)	4	–1	10	20
Total				30	100

MATHEMATICS

Section	Question Type	+Ve Marks	–Ve Marks	No. of Qs.	Total marks
Sec–I (Q.N : 61 – 80)	Questions with Single Correct Choice	4	–1	20	80
Sec–II (Q.N : 81 – 90)	Questions with Numerical Answer Type (+ / – Number)	4	–1	10	20
Total				30	100

Exam Syllabus

PHYSICS	Gravitation, Mechanical Properties of Solids, Mechanical Properties of Fluid, Thermal Properties of Matter, Thermodynamics, Kinetic Theory of Gases Oscillations Waves
CHEMISTRY	Equilibrium, Some p-block elements, Organic Chemistry : Some Basic Principle and Techniques, Hydrocarbon
MATHEMATICS	Straight Line, Circles Complex number Conic Section–II, Properties of Triangles, Statistics

PHYSICS

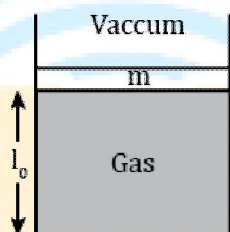
Max. Marks: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

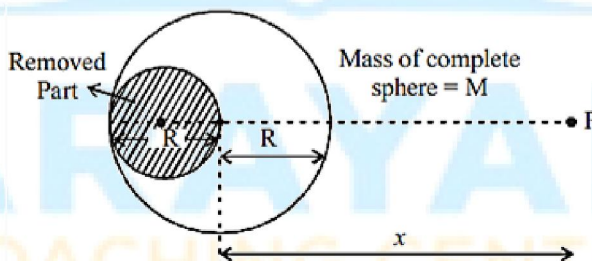
This section contains 20 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

1. A smooth piston of mass m and cross-section area A is in equilibrium on a diatomic gas filled in a cylindrical container and outside there is vacuum. The piston and the container are non-conducting and the height of gas column in the container is ℓ_0 at equilibrium. If the piston is displaced slightly down the equilibrium position, then the time period of its oscillation will be:



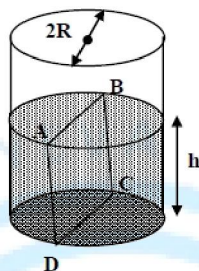
- A) $2\pi\sqrt{\frac{10\ell_0}{7g}}$ B) $2\pi\sqrt{\frac{7\ell_0}{10g}}$ C) $2\pi\sqrt{\frac{7\ell_0}{5g}}$ D) $2\pi\sqrt{\frac{5\ell_0}{7g}}$
2. The gravitational field, due to the 'leftover part' of a uniform sphere (from which a part as shown, has been 'removed out') at a very far off point, P, located as shown, would be (nearly):



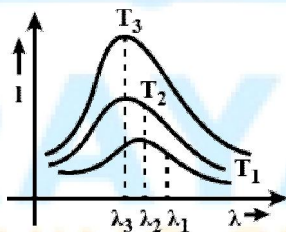
- A) $\frac{5}{6} \frac{GM}{x^2}$ B) $\frac{8}{9} \frac{GM}{x^2}$ C) $\frac{7}{8} \frac{GM}{x^2}$ D) $\frac{6}{7} \frac{GM}{x^2}$

Rough Work

3. Water is filled up to a height h in a beaker of radius R as shown in the figure. The density of water is ρ , the surface tension of water is T and the atmospheric pressure is P_0 . Consider a vertical section ABCD of the water column through a diameter of the beaker. The force on water on one side of this section by water on the other side of this section has magnitude



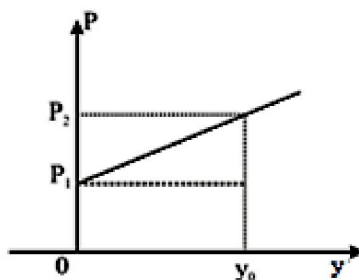
- A) $|2P_0Rh + \pi R^2 \rho gh - 2RT|$ B) $|2P_0Rh + R\rho gh^2 - 2RT|$
 C) $|P_0\pi R^2 + R\rho gh^2 - 2RT|$ D) $|P_0\pi R^2 + R\rho gh^2 + 2RT|$
4. In an energy recycling process, 100g of steam at 100°C becomes water at 100°C . Which converts y g of ice at 0°C into water at 100°C . The number value of y is
 A) 100 B) 200 C) 300 D) 400
5. According to Wien's displacement law:



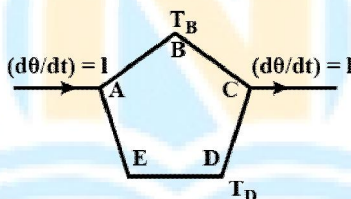
- A) Total area under the three graphs A_1, A_2, A_3 are equal
 B) Areas are in the ratio: $A_1 : A_2 : A_3 = T_1^2 : T_2^2 : T_3^2$
 C) Areas are in the ratio: $A_1 : A_2 : A_3 = T_1^4 : T_2^4 : T_3^4$
 D) $T_3 < T_2 < T_1$

Rough Work

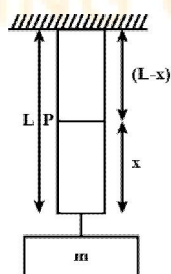
6. The pressure of liquid varies with depth 'y' as shown in figure. Then the density of the liquid is



- A) $\rho = \frac{P_2 P_1}{g y_0}$ B) $\rho = \frac{P_2 - P_1}{g y_0}$ C) $\rho = \frac{P_2 + P_1}{g y_0}$ D) $\rho = \frac{P_2 - P_1}{2 g y_0}$
7. A mass of 50 kg is placed at the centre of a uniform spherical shell of mass 100 kg and radius 50 m. If the gravitational potential at a point, 25m from the centre is V. The value of V is:
A) -60G B) +2G C) -20G D) -4G
8. ABCDE is a regular pentagon of uniform wire. The rate of heat entering at A and leaving at C is equal. T_B and T_D are temperature of B and D. Find the temperature T_C .



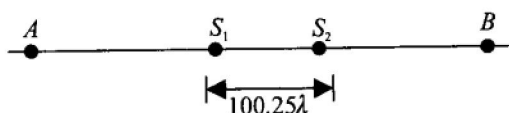
- A) $\frac{3T_B + 2T_D}{5}$ B) $3T_D - 2T_B$ C) $3T_D + 2T_B$ D) can have any value
9. One end of a uniform wire of length 'L' and mass 'M' is attached rigidly to a point suspended from its lower end, A block of mass m is also hanging from free end of the wire. If a is the area of cross-section of the wire, then the stress in the wire at height $x = \frac{L}{3}$ from its lower end is



- A) $\frac{Mg}{a} + \frac{mg}{3a}$ B) $\frac{mg}{a} - \frac{Mg}{3a}$ C) $\frac{mg}{a} + \frac{Mg}{3a}$ D) $\frac{mg}{aL} + \frac{MLg}{3a}$

Rough Work

10. A lead bullet strikes against a steel plate with a velocity 200ms^{-1} . If the impact is perfectly inelastic and the heat produced is equally shared between the bullet and the target, then the rise in temperature of the bullet is (specific heat capacity of lead = $125\text{Jkg}^{-1}\text{K}^{-1}$)
- A) 80°C B) 60°C C) 160°C D) 40°C
11. Three perfect gases at absolute temperatures T_1, T_2 and T_3 are mixed. The mass of molecules are m_1, m_2 and m_3 and the number of molecules are n_1, n_2 and n_3 respectively. Assuming no loss of energy, the final temperature of the mixture is:
- A) $\frac{n_1T_1 + n_2T_2 + n_3T_3}{n_1 + n_2 + n_3}$ B) $\frac{n_1T_1 + n_2T_2^2 + n_3T_3^2}{n_1T_1 + n_2T_2 + n_3T_3}$
- C) $\frac{n_1^2T_1^2 + n_2^2T_2^2 + n_3^2T_3^2}{n_1T_1 + n_2T_2 + n_3T_3}$ D) $\frac{(T_1 + T_2 + T_3)}{3}$
12. The equation of state for a gas is given by $PV = nRT + \alpha V$, where n is the number of moles and α is a positive constant. The initial temperature and pressure of one mole of the gas contained in a cylinder are T_o and P_o respectively. The work done by the gas when its temperature doubles isobarically will be:
- A) $\frac{P_o T_o R}{P_o - \alpha}$ B) $\frac{P_o T_o R}{P_o + \alpha}$ C) $P_o T_o R$ in 2 D) $P_o T_o R$
13. S_1 and S_2 are two coherent sources of sound separated by distance 100.25λ , where λ is the wave length of sound. S_1 leads S_2 in phase by $\pi/2$. A and B are two points on the line joining S_1 and S_2 as shown in the figure. The ratio of amplitudes of sources S_1 and S_2 are in ratio 1:2. The ratio of intensity at A to that B $\left(\frac{I_A}{I_B}\right)$ is:



- A) ∞ B) $1/9$ C) 0 D) 9

Rough Work

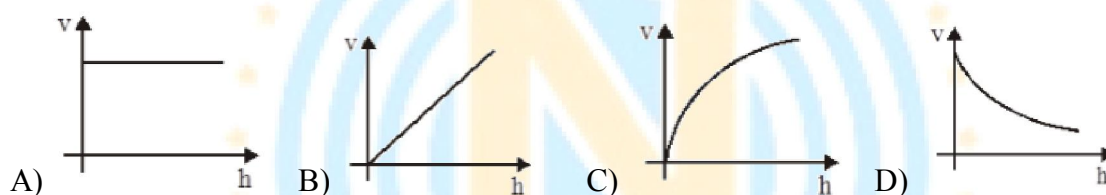
14. Two particles P and Q describe S.H.M. of same amplitude a , same frequency f along the same straight line from the same mean position. The maximum distance between the two particles is $a\sqrt{2}$. The phase difference between the particle is:

A) zero B) $\pi/2$ C) $\pi/8$ D) $\pi/3$

15. A horizontal spring-block system of mass 2kg executes S.H.M. When the block is passing through equilibrium position, an object of mass 1kg is put on it and the two move together. The new amplitude of vibration is (A being its initial amplitude):

A) $\sqrt{\frac{2}{3}}A$ B) $\sqrt{\frac{3}{2}}A$ C) $\sqrt{2}A$ D) $\frac{A}{\sqrt{2}}$

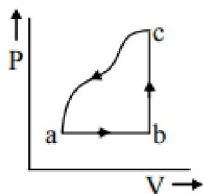
16. A uniform rope having mass m hangs vertically from a rigid support. A transverse wave pulse is produced at the lower end. The speed u of wave pulse varies with height h from the lower end as:



17. Two materials coefficients of thermal conductivity ' $3K$ ' and ' K ' and thickness ' d ' and ' $3d$ ', respectively, are joined to form a slab as shown in the figure. The temperature of the outer surfaces are ' θ_2 ' and ' θ_1 ' respectively, ($\theta_2 > \theta_1$). The temperature at the interface is :

A) $\frac{\theta_1}{10} + \frac{9\theta_2}{10}$ B) $\frac{\theta_2 + \theta_1}{2}$ C) $\frac{\theta_1}{6} + \frac{5\theta_2}{6}$ D) $\frac{\theta_1}{3} + \frac{2\theta_2}{3}$

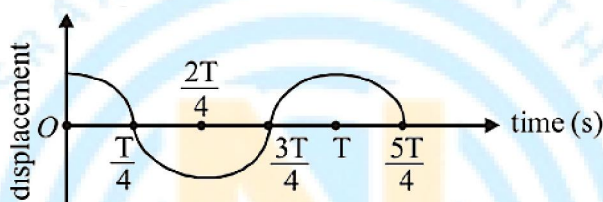
18. A sample of an ideal gas is taken through the cyclic process abca as shown in figure. The change in the internal energy of the gas along the path ca is -180J , The gas absorbs 250 J of heat along the path ab and 60 J along the path ac. The work done by the gas along the path abc is:



A) 120 J B) 130 J C) 100 J D) 140 J

Rough Work

19. The temperature, at which the root mean square velocity of hydrogen molecules equals their escape velocity from the earth, is closest to
[Boltzmann constant $k_B = 1.38 \times 10^{-23}$ J/K
Avogadro Number $N_A = 6.02 \times 10^{26}$ /kg
Radius of earth : 6.4×10^6 m
Gravitational acceleration on Earth = 10 ms^{-2}]
- A) 800 K B) 3×10^5 K C) 10^4 K D) 650 K
20. The displacement time graph of a particle executing SHM is given in figure: (sketch is schematic and not to scale)



Which of the following statements is/are true for this motion?

- (1) The force is zero at $t = \frac{3T}{4}$
 (2) The acceleration is maximum at $t = T$
 (3) The speed is maximum at $t = T/4$
 (4) The P.E. is equal to K.E. of the oscillation at $t = \frac{T}{2}$
- A) (1), (2) and (4) B) (2), (3) and (4)
 C) (1), (2) and (3) D) (1) and (4)

Rough Work

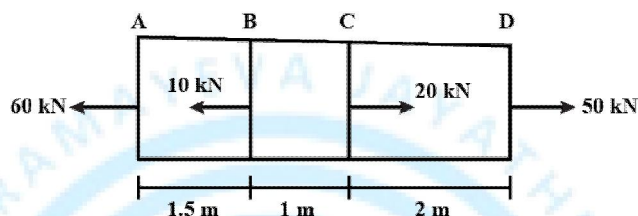
SECTION-II

(Numerical Value Answer Type)

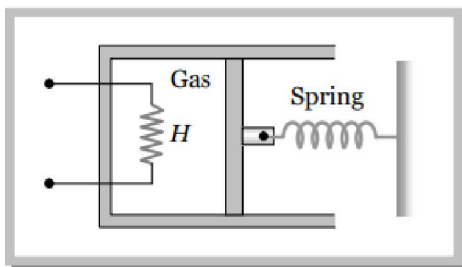
This section contains 10 questions. The answer to each question is a Numerical value. If the Answer is in the decimals, **Mark the nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

21. A steel rod of cross-sectional area 2 m^2 is acted upon by forces as shown in fig. The total elongation of the bar is $P \times 10^{-7}\text{ m}$. Compute the value of P. (Take $Y = 2.0 \times 10^{11}\text{ N/m}^2$).



22. 2 kg of monoatomic gas is at a pressure of $4 \times 10^4\text{ N/m}^2$. The density of the gas is 8 kg/m^3 . The energy $n \times 10^3\text{ J}$ of the gas due to its thermal motion. The value of n?
23. If one wants to remove all the mass of the earth to infinity in order to break it up completely. The amount of energy that needs to be supplied will be $\frac{x}{5} \frac{Gm^2}{R}$. Where x is _____ (Round off to the nearest Integer).
(M is the mass of the earth, R is the radius of earth, G is the Gravitational constant).
24. An ideal mono-atomic gas is confined in a cylinder by a spring loaded piston of cross section $0.8 \times 10^{-2}\text{ m}^2$. Initially the gas is at 300 K and occupies a volume of $2.4 \times 10^{-3}\text{ m}^3$ and the spring is in its relaxed state as shown in figure. The gas is heated by a small heater until the piston moves out slowly by 0.1 m . The force constant of the spring is 8000 N/m and the atmospheric pressure is $1.0 \times 10^5\text{ N/m}^2$. The cylinder and the piston are thermally insulated. The piston and the spring are massless and there is no friction between the piston and the cylinder. The final temperature of the gas will be in K : (Neglect the heat loss through the lead wires of the heater. The heat capacity of the heater coil is also negligible):



Rough Work

25. C_p and C_v are specific heats at constant pressure and constant volume respectively. It is observed that:
- $C_p - C_v = a$ for hydrogen gas
- $C_p - C_v = b$ for nitrogen gas
- Then find out a / b is:
26. If a capillary tube of radius 1 mm is immersed in water, the mass of water rising in the capillary tube is M . If the radius tube is doubled, then the mass of water, that rises in the capillary tube is λM , find the value of λ
27. Two waves are simultaneously passing through a string and their equations are :
 $y_1 = A_1 \sin k(x - vt)$, $y_2 = A_2 \sin k(x - vt + x_0)$. Given amplitudes $A_1 = 12\text{mm}$ and $A_2 = 5\text{mm}$, $x_0 = 3.5\text{ cm}$ and wave number $k = 6.28\text{ cm}^{-1}$. The amplitude of resulting wave will be ____ mm.
28. The initial velocity v_i required to project a body vertically upwards from the surface of the earth to reach a height of $10R$, where R is the radius of the earth. v_i can be described in terms of escape velocity v_e as $v_i = \sqrt{\frac{x}{11}} \times v_e$. the value of x will be _____
29. The amplitude of a simple harmonic oscillator is 2 cm. At certain instant its potential energy and kinetic energy are equal. Displacement from mean position at this instant in cm is $\sqrt{x}$ cm. Find out x .
30. The gravitational field in a region is given by $\vec{g} = 5\text{N/kg}\hat{i} + 12\text{N/kg}\hat{j}$. The change in the gravitational potential energy of a particle of mass 1kg when it is taken from the origin to a point $(7\text{m}, -3\text{m})$ is:

Rough Work

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options A), B), C) and D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

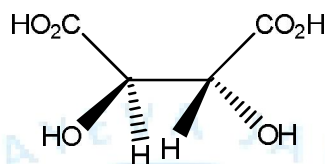
31. In HS^- , I^- , $\text{R}-\text{NH}_2$, NH_3 order of proton accepting tendency will be:
- A) $\text{I}^- > \text{NH}_3 > \text{RNH}_2 > \text{HS}^-$ B) $\text{NH}_3 > \text{RNH}_2 > \text{HS}^- > \text{I}^-$
 C) $\text{RNH}_2 > \text{NH}_3 > \text{HS}^- > \text{I}^-$ D) $\text{HS}^- > \text{RNH}_2 > \text{NH}_3 > \text{I}^-$
32. Assuming complete dissociation which of the following aqueous solutions will have the same pH value?
- (i) 100 mL of 0.01 M HCl
 (ii) 100 mL of 0.01 M H_2SO_4
 (iii) 50 mL of 0.01 M HCl
 (iv) Mixture of 50 mL of 0.02 M H_2SO_4 and 50 mL of 0.02 M NaOH
- A) (i), (ii) B) (i), (iii) C) (ii), (iv) D) (i), (iv)
33. pH for the solution of salt undergoing anionic hydrolysis (say CH_3COONa) is given by:
- A) $\text{pH} = \frac{1}{2}[\text{pK}_w + \text{pK}_a + \log c]$ B) $\text{pH} = \frac{1}{2}[\text{pK}_w + \text{pK}_a - \log c]$
 C) $\text{pH} = \frac{1}{2}[\text{pK}_w + \text{pK}_b - \log c]$ D) None of the above
34. The concentration of oxalic acid is 'x' mol L^{-1} , if 40 mL of this solution reacts with 16 mL of 0.05 M acidified KMnO_4 . What is the pH of 'x' M oxalic acid solution?
 (Assume that oxalic acid dissociates completely)
- A) 1.3 B) 1.699 C) 1 D) 2
35. Which statement is false? (Assume complete dissociation in each case)
- A) If 2.0 L of a solution of H_2SO_4 contains 0.1 mole, then pH of the solution is 1
 B) The concentration of OH^- in 0.005 M HNO_3 is $2.0 \times 10^{-12} \text{ mol/L}$
 C) The pH of 0.01 M KOH is 12
 D) In a 0.001 M solution of NaOH the concentration of H^+ is 10^{-3} mol/L

Rough Work

36. 10 mL of a solution contains 0.1 M NH_4Cl + 0.01 M NH_4OH . Which addition would not change the pH of the solution?

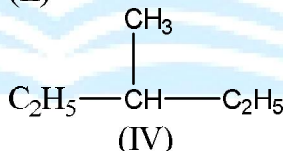
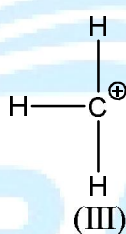
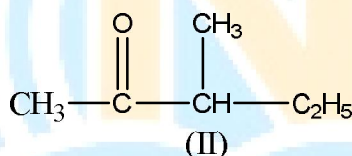
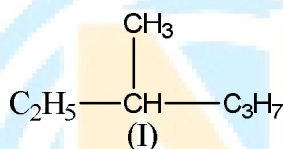
- A) Adding 1mL water
B) Adding 5 mL of 0.1M NH_4Cl
C) Adding 5mL of 0.1M NH_4OH
D) Adding 10mL of 0.1M NH_4Cl

37. The absolute configuration of



- A) S, S
B) R, R
C) R, S
D) S, R

38. Among the following four structures I to IV



it is true that

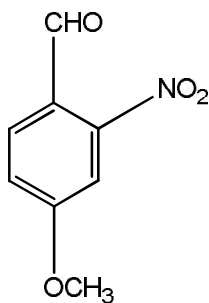
- A) All four are chiral compounds
B) Only I and II are chiral compounds
C) Only III is a chiral compound
D) Only II and IV are chiral compounds

39. The Beilstein test for organic compound is used to detect

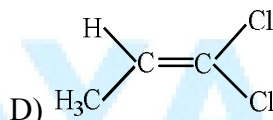
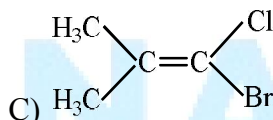
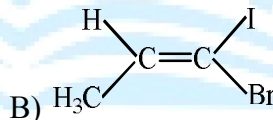
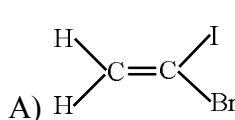
- A) Nitrogen
B) Sulphur
C) Carbon
D) Halogens

Rough Work

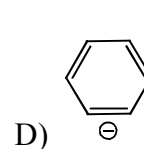
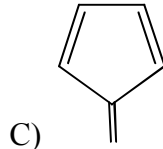
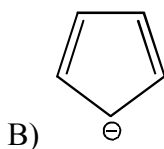
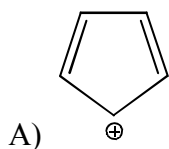
40. What is the correct IUPAC name of



- A) 4-methoxy-2-nitrobenzaldehyde B) 4-formyl-3-nitro anisole
C) 4-methoxy-6-nitrobenzaldehyde D) 2-formyl-5-methoxy nitrobenzene
41. Which of the following orders is correct regarding the $-I$ effect of the substituents?
- A) $-NR_2 > -OR > -F$ B) $-NR_2 < -OR < -F$
C) $-NR_2 > -OR < -F$ D) $-NH_2 < -OR > -F$
42. In Kjeldahl's method of estimation of nitrogen, $CuSO_4$ acts as
- A) Oxidising agent B) Reducing agent
C) Catalytic agent D) Hydrolysis agent
43. Geometrical isomerism is shown by :

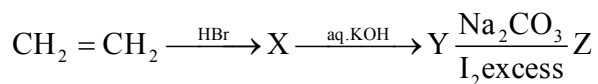


44. An alkene on reductive ozonolysis gives 2-molecules of $CH_2(CHO)_2$. The alkene is
- A) 2,4-hexadiene B) 1,3-cyclohexadiene
C) 1,4-cyclohexadiene D) 1-methyl-1, 3-cyclopentadiene
45. Which of the following species is aromatic?



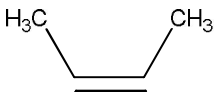
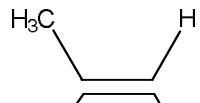
Rough Work

46. Identify Z in the series,

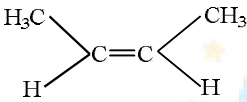


- A) $\text{C}_2\text{H}_5\text{I}$ B) $\text{C}_2\text{H}_5\text{OH}$ C) CHI_3 D) CH_3CHO

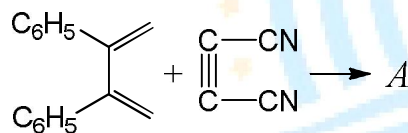
47. The major product in the reaction of 2-butyne with Li/liq. NH_3 is

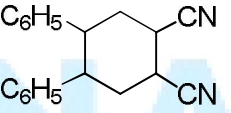
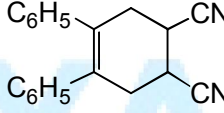
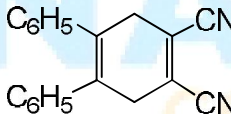
- A)  B) 
C) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ D) $\text{H}_2\text{C} = \text{CH} - \text{CH}_2 - \text{CH}_3$

48. Which of the following hydrocarbons has the lowest dipole moment?

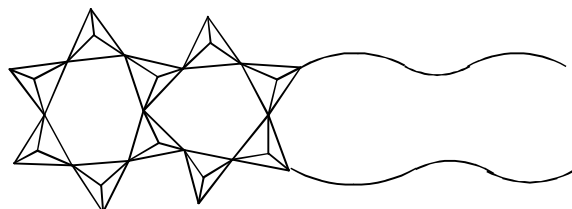
- A)  B) $\text{CH}_3\text{C} \equiv \text{CCH}_3$
C) $\text{CH}_3\text{CH}_2\text{C} \equiv \text{CH}$ D) $\text{CH}_2 = \text{CH} - \text{C} \equiv \text{CH}$

49. Identify A:



- A)  B) 
C)  D) None of these

50. Silicate structure unit of



- A) $(\text{Si}_4\text{O}_{11})_n^{-6n}$ B) $(\text{Si}_2\text{O}_{11})_n^{-2n}$ C) (Si_2O_3) D) $(\text{SiO}_4)^{-4}$

Rough Work

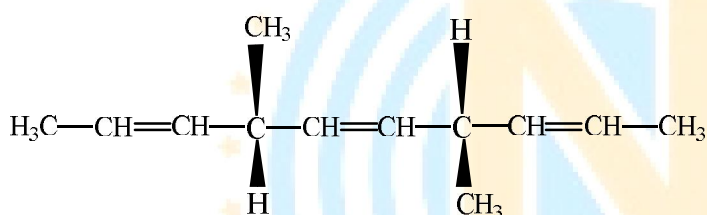
SECTION-II

(Numerical Value Answer Type)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer is in the decimals, **Mark the nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

51. If the concentration of OH^- ions is the reaction $\text{Fe}(\text{OH})_3(s) \rightleftharpoons \text{Fe}^{3+}(aq) + 3\text{OH}^-(aq)$, is decreased by $\frac{1}{4}$ times, then equilibrium concentration of Fe^{3+} will increase by :
52. One mole of ethyl alcohol was treated with one mole of acetic acid at 25°C . $\frac{2}{3}$ of the acid changes into ester at equilibrium. The equilibrium constant for the reaction will be:
53. The number of optically active products obtained from the complete ozonolysis of the given compound is :



54. Number of acidic hydrogen atom in butyne-1 is:
55. The number of possible isomers of alkane with formula C_6H_{14} is:
56. How many stereoisomers does this molecule have?
 $\text{CH}_3\text{CH}=\text{CHCH}_2\text{CHBrCH}_3$
57. Benzaldoxime exists in how many forms?
58. In aluminates coordination number of Al is:
59. The mass of carbon anode consumed (giving only carbon dioxide) in production of 270 kg of aluminium metal from bauxite by the Hall process is (Atomic mass of Al=27)
60. Which of the following having +M Groups.
 OH , NH_2 , Cl , CHO , COOH , NO_2 , CN , OCH_3

Rough Work

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options A), B), C) and D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

61. The normal at a variable point P on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$) of eccentricity 'e' meets the axis of the ellipse at Q and R. Then the locus of the mid-point of the QR is a conic with eccentricity ' e_1 ' such that
- A) e_1 is independent of e
B) $e_1 = 1$
C) $e_1 = e$
D) $e_1 = \frac{1}{e}$
62. The equation of the common tangent touching the circle $(x-3)^2 + y^2 = 9$ and the parabola $y^2 = 4x$ above the x-axis is
- A) $\sqrt{3}y = 3x + 1$
B) $\sqrt{3}y = -(x + 3)$
C) $\sqrt{3}y = x + 3$
D) $\sqrt{3}y = -(3x + 1)$
63. If 'z' lies on the circle $|z - 2i| = 2\sqrt{2}$, then the value of $\arg\left(\frac{z-2}{z+2}\right)$ is equal to
- A) $\frac{\pi}{3}$
B) $\frac{\pi}{4}$
C) $\frac{\pi}{6}$
D) $\frac{\pi}{2}$
64. The latusrectum subtends a right angle at the centre of the ellipse, then its eccentricity is
- A) $2 \sin 18^\circ$
B) $2 \cos 18^\circ$
C) $2 \sin 54^\circ$
D) $2 \cos 54^\circ$
65. Let $P(4, -4)$ and $Q(9, 6)$ be two points on the parabola $y^2 = 4x$ and let X be any point on the arc POQ of this parabola, where O is vertex of the parabola, such that the area of ΔPXQ is maximum, then its maximum area (in sq units) is
- A) $\frac{75}{2}$
B) $\frac{125}{4}$
C) $\frac{625}{4}$
D) $\frac{125}{2}$

Rough Work

66. Tangents drawn from the point $P(1,8)$ to the circle $x^2 + y^2 - 6x - 4y - 11 = 0$ touch the circle at the points A and B. The equation of the circumcircle of the triangle in PAB is
- A) $x^2 + y^2 + 4x - 6y + 19 = 0$ B) $x^2 + y^2 - 4x - 10y + 19 = 0$
 C) $x^2 + y^2 - 2x + 6y - 29 = 0$ D) $x^2 + y^2 - 6x - 4y + 19 = 0$
67. The length of the transverse axis along x -axis with centre at origin of a hyperbola is 7 and it passes through the point $(5, -2)$ The equation of the hyperbola is
- A) $\frac{4}{49}x^2 - \frac{196}{51}y^2 = 1$ B) $\frac{49}{4}x^2 - \frac{51}{196}y^2 = 1$
 C) $\frac{4}{49}x^2 - \frac{51}{196}y^2 = 1$ D) None
68. Let $P(3, 3)$ be a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If the normal to it at P intersects the x -axis at $(9, 0)$ and e is its eccentricity, then the ordered pair (a^2, e^2) is equal to
- A) $\left(\frac{9}{2}, 3\right)$ B) $\left(\frac{3}{2}, 2\right)$ C) $\left(\frac{9}{2}, 2\right)$ D) $(9, 3)$
69. Let k be an integer such that the triangle with vertices $(k, -3k)$, $(5, k)$ and $(-k, 2)$ has area 28 sq. units. Then the orthocentre of this triangle is at the point (α, β) then $\alpha\beta$
- A) 1 B) 2 C) 3 D) 4
70. Let $0 < \theta < \frac{\pi}{2}$, if the eccentricity of the hyperbola $\frac{x^2}{\cos^2 \theta} - \frac{y^2}{\sin^2 \theta} = 1$ is greater than 2, then the length of its latus rectum lies in the interval
- A) $(3, \infty)$ B) $\left(\frac{3}{2}, 2\right]$ C) $(2, 3]$ D) $\left(1, \frac{3}{2}\right]$

Rough Work

71. Locus of the vertices of an equilateral triangle circumscribing the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is
- A) $3(x^2 + y^2 - a^2 - b^2)^2 = 4(b^2x^2 + a^2y^2 - a^2b^2)$
- B) $(x^2 + y^2 - a^2 - b^2)^2 = (b^2x^2 + a^2y^2 - a^2b^2)$
- C) $(x^2 + y^2 - a^2 - b^2)^2 = 4(b^2x^2 + a^2y^2 - a^2b^2)$
- D) $4(x^2 + y^2 - a^2 - b^2)^2 = 3(b^2x^2 + a^2y^2 - a^2b^2)$
72. If $iz^3 + z^2 - z + i = 0$, then $|z| =$
- A) 1 B) 2 C) -1 D) -2
73. Suppose that the normal drawn at three different points on the parabola $y^2 = 4x$ pass through the point $(h, 0)$ then the value of "h" is greater than
- A) 0 B) -4 C) 2 D) 3
74. Locus of the mid-points of the chord of contact of tangents drawn from the point lying on the straight line $4x - 5y = 20$ to the circle $x^2 + y^2 = 9$ is
- A) $20(x^2 + y^2) - 36x + 45y = 0$ B) $20(x^2 + y^2) + 36x - 45y = 0$
- C) $36(x^2 + y^2) - 20x + 45y = 0$ D) $36(x^2 + y^2) + 20x - 45y = 0$
75. The mean and S.D. of the marks of 200 candidates were found to be 40 and 15 respectively. Later, it was discovered that a score of 40 was wrongly read as 50. The correct mean and S.D. respectively are _____.
- A) (14.98, 40.40) B) (39.95, 14.98) C) (39.95, 15.98) D) (14.98, 39.95)
76. A line meets the coordinate axes in A and B, A circle is circumscribed about the triangle OAB. If m and n are the distances of the tangent to the circle at the origin from the points A and B respectively, the diameter of the circle is
- A) $m(m+n)$ B) $m+n$ C) $n(m+n)$ D) $\left(\frac{1}{2}\right)(m+n)$

Rough Work

77. A straight line through a fixed point (2,3) intersects the coordinate axes at distinct points P and Q. If O is the origin and the rectangle OPRQ is completed, then the locus of R is :
- A) $3x + 2y = 6$ B) $2x + 3y = xy$ C) $3x + 2y = xy$ D) $3x + 2y = 6xy$
78. Which of the following line is concurrent with the lines $3x + 4y + 6 = 0$ and $6x + 5y + 9 = 0$
- A) $2x + 3y + 5 = 0$ B) $3x + 3y + 5 = 0$
 C) $7x + 9y + 3 = 0$ D) $9x + 7y + 11 = 0$
79. $A(z_1), B(z_2), C(z_3)$ are the vertices of an equilateral triangle ABC, the value of $\arg\left(\frac{z_2 + z_3 - 2z_1}{z_3 - z_2}\right)$ is equal to
- A) $\frac{\pi}{4}$ B) $\frac{\pi}{2}$ C) $\frac{\pi}{3}$ D) $\frac{\pi}{6}$
80. If z_1, z_2 are roots of equation $z^2 - az + a^2 = 0$, then $\left|\frac{z_1}{z_2}\right| =$
- A) a^2 B) a C) $2a$ D) 1

SECTION-II (Numerical Value Answer Type)

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81. Number of all non-zero complex no's Z satisfying $\bar{z} = iz^2$
82. A bridge is in the shape of a semi-ellipse. It is 400mts long and has a maximum height 10mts at the middle point. The height of the bridge at a point which is at a distance 80mts from one end is _____ mts
83. The shortest distance between the parabolas $y^2 = x - 1$ and $x^2 = y - 1$ is $\frac{3\sqrt{2}}{k}$ then the value of numerical quantity " k " must be
84. The locus of the point of intersection of the lines $\sqrt{3}kx + ky - 4\sqrt{3} = 0$ and $\sqrt{3}x - y - 4\sqrt{3}k = 0$ is a conic, whose eccentricity is

Rough Work

85. The average weight of students in a class of 35 students is 40 kg. If the weight of the teacher is included, the average rises by $\frac{1}{2}$ kg; the weight of the teacher is _____
86. The perpendicular bisector of the line segment joining P(1, 4) and Q(k,3) has y-intercept -4. Then, a possible value of |k| is
87. The distance of the line $2x - 3y = 4$ from the point (1, 1) measured parallel to the line $x + y = 1$ is d then the value of $d\sqrt{2}$
88. P and Q are two distinct points on the parabola $y^2 = 4x$ with parameters t and t_1 respectively. If the normal at P passes through Q, then the minimum value of t_1^2 is
89. Runs scored by a batsman in 10 innings are: 38, 70, 48, 34, 42, 55, 63, 46, 54, 44 the mean deviation is _____.
90. Two circles are inscribed and circumscribed about a square ABCD, length of each side of the square is 32. P and Q are two points respectively on these circles, then $\frac{1}{16} \left[\sum (QA)^2 - \sum (PA)^2 \right]$ is equal to

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